Lab 4: Connect Data Agent to an AI Search Index

Overview

In this lab, you will build a Fabric Data Agent that answers questions using the ecommerce-dataset semantic model and connects to **Azure AI Search index** created in **Azure AI Foundry**, enabling the data agent to reason over **unstructured/semi-structured** such as PDFs, JSON and text files. You'll start with creating a connection to an already provisioned AI search index in your data agent and configuring it to reason over customer reviews for the ecommerce products. Finally, you'll test the agent with sample questions on customer reviews.

Prerequisites

1. You have completed Lab 1 to Lab 3 exercises.

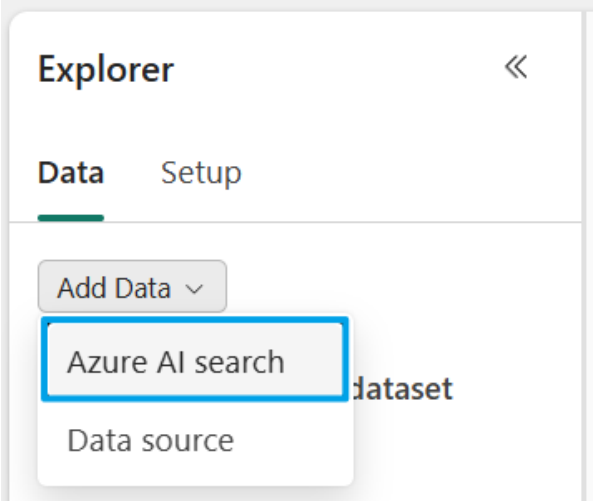
Learning Objectives

1. Adding an AI Search Index connection in your Data Agent.
2. Configuring Agent Instructions and source context to the AI search index connection that helps data agent route and answer questions correctly.

Lab Steps

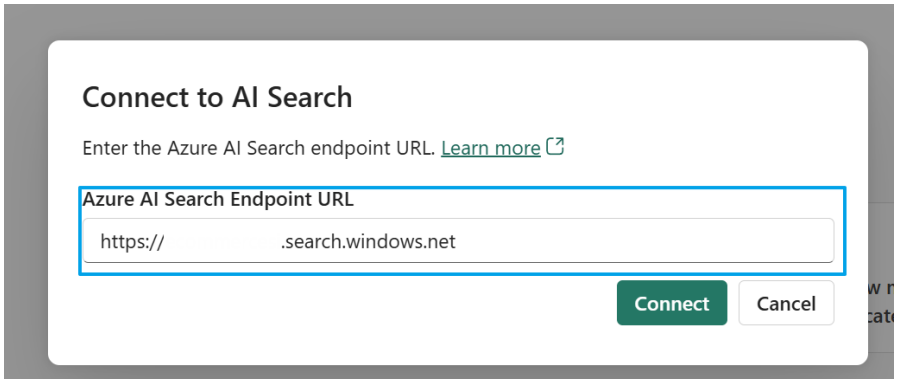
Step 1: Connect AI Search Index to your Data Agent

1. Open the data agent in your workspace that you created in Lab 2. On the explorer pane Click "Add Data" and select "+ Azure AI Search" .

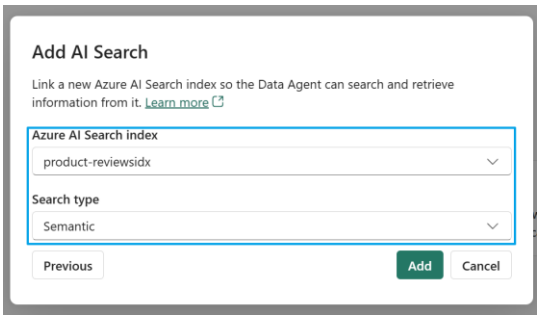


2. Add the provided Azure AI Search endpoint in the next dialog box and click connect . AI Search endpoint:

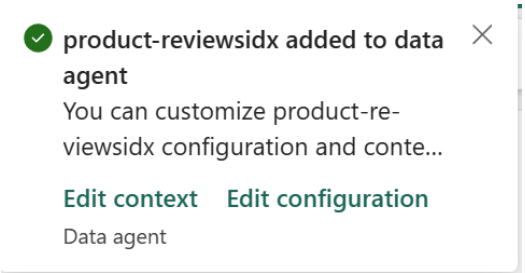
https://ecommercess.search.windows.net



3. Select "***product-reviewsidx***" as the search index and "**Semantic**" as the search type in the dialog box and click add.



The connection will be validated and you'll get a notification –"product-reviewsidx added to data agent".

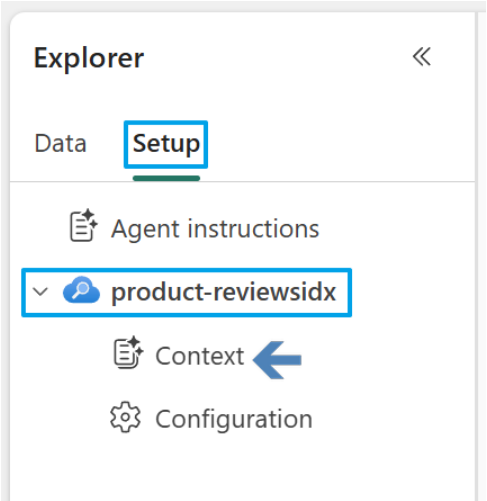


Step 2: Configure Agent Instructions and add context to the AI search index connection

You can configure how your Data Agent uses your AI Search index to enhance its ability to answer questions accurately and efficiently. This involves providing context about the source, specifying which fields should be referenced, and adjusting parameters that determine how much context is returned with responses.

In this step we will add agent instructions and data source context to ensure our data agent uses the AI Search Index accurately based on our requirement.

1. On the explorer pane click on “Setup” and select “product-reviewsidx” and then “Context” to open editor UI for the agent.

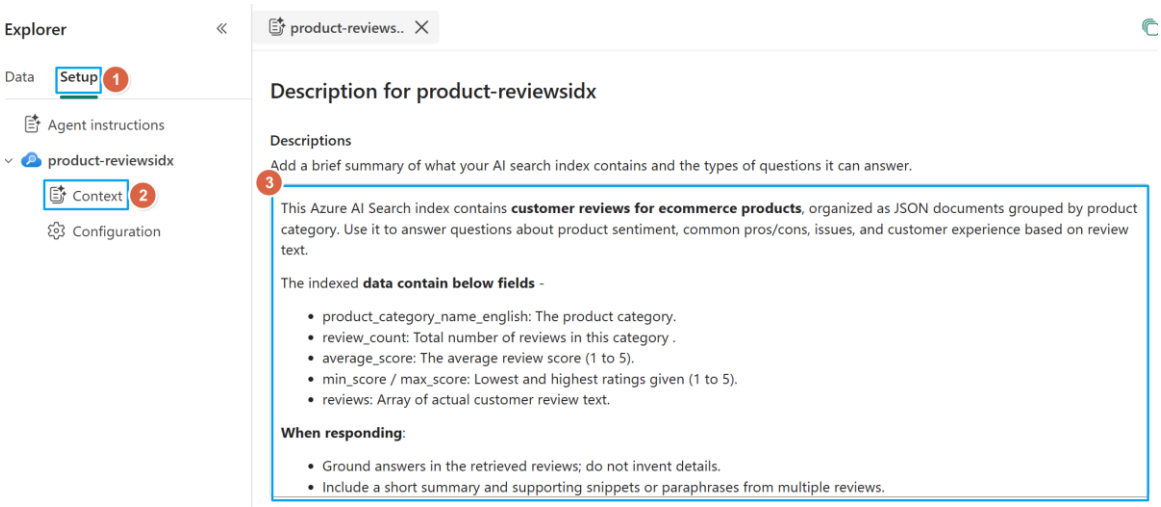


2. In the Description section add below text to describe the AI Search index

This Azure AI Search index contains **customer reviews for ecommerce products**, organized as JSON documents grouped by product category. Use it

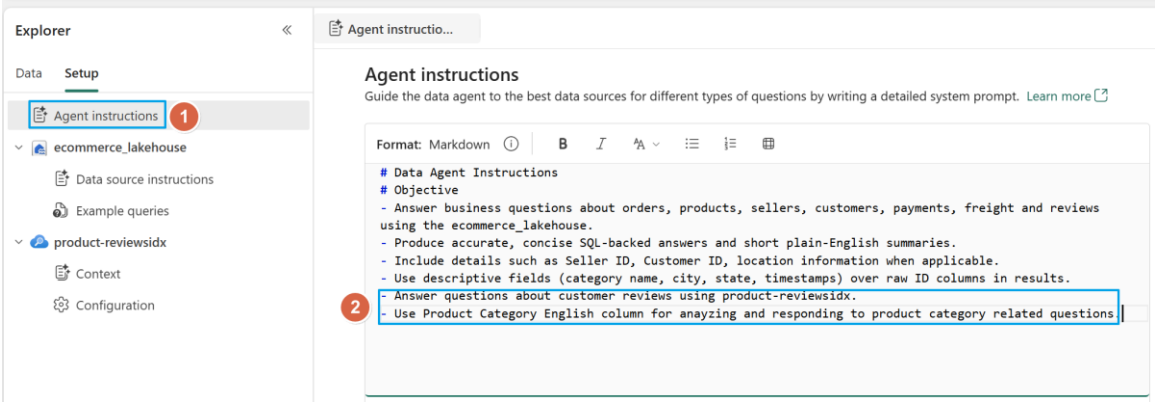
to answer questions about product sentiment, common pros/cons, issues, and customer experience based on review text. The indexed **data contain below fields** -

- product_category_name_english: The product category.
- review_count: Total number of reviews in this category .
- average_score: The average review score (1 to 5).
- min_score / max_score: Lowest and highest ratings given (1 to 5).
- reviews: Array of actual customer review text.



3. Under "Agent instructions" add below two lines at the end of agent instructions markdown:

- Answer questions about customer reviews using product-reviewsidx.
- Use Product Category English column for anayzing and responding to product category related questions.

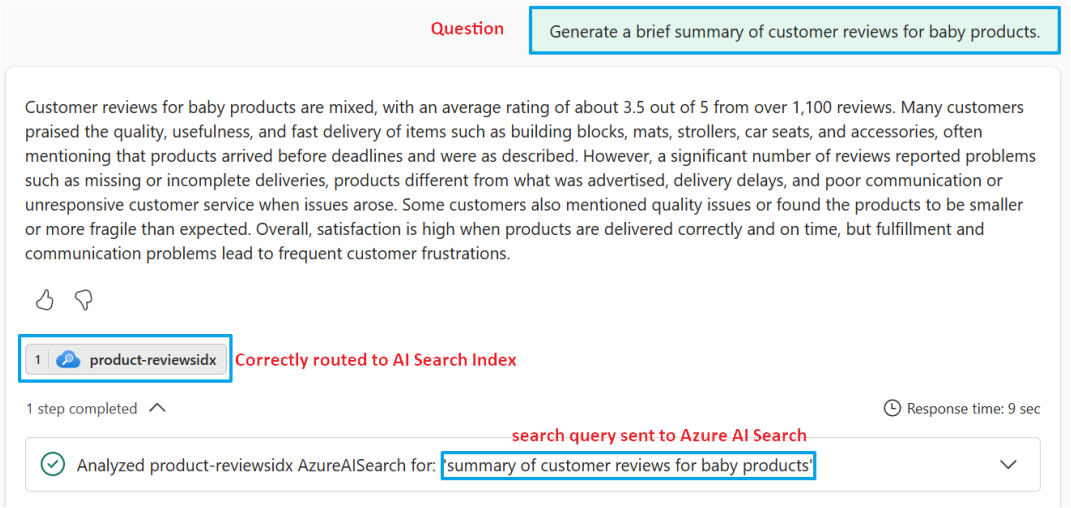


4. Publish the data agent with the added AI Search Index context and updated

agent instructions.

Step 3: Test and validate

1. In the "Test the agent's responses" pane, ask the question – "***What are top reviews for baby products?***"



You can observe from the response that the data agent routed the question correctly and generated a search query to be sent to Azure AI search index.

2. Next let's try asking "***What are the common problems reported by customers in reviews for electronics?***"

What are the common problems reported by customers in reviews for electronics?

Based on customer reviews for electronics, the most common problems reported are:

1. Defective Products: Many customers mention receiving electronics that do not work, arrive broken, or stop functioning soon after use (e.g., "the device arrived with a defect," "the product does not work").
2. Incorrect or Missing Items: A frequent issue is receiving the wrong product, a counterfeit or generic alternative, or incomplete orders (e.g., "I received only one device," "the product is not as shown in the image," "received fake remotes").
3. Delivery Problems: Customers often experience delayed deliveries, partial deliveries, or orders never arriving (e.g., "product delayed," "I have not received the product yet," "delivery took very long").
4. Poor Customer Service: Some customers struggle to get responses when problems occur, especially when requesting exchanges or refunds (e.g., "unable to reach customer service," "no response about missing items").
5. Quality Issues: Reports include fragile materials, cosmetic damage during shipping, or products not matching descriptions in terms of build or features (e.g., "very fragile," "quality of the product leaves much to be desired").
6. Packaging and Accessories: Items sometimes arrive with damaged packaging, missing manuals, or incompatible batteries/accessories (e.g., "packaging looks like it was taken from a dump," "batteries not compatible," "lacked instruction manual").

Overall, the most common complaints focus on defective or incorrect products, delayed or missing deliveries, and challenges getting support for these issues after purchase.

1 product-reviewsidx

1 step completed

Response time: 17 sec

Analyzed product-reviewsidx AzureAISearch for: "common problems reported in customer reviews for electronics category"

If you expand the step competed text box, you can view the JSON output from Azure AI search endpoint containing the response and additional metadata for the search.

Analyzed product-reviewsidx AzureAISearch for: "common problems reported in customer reviews for electronics category"

Document 1

Json


```
{
  "id": "aHR0cHM6Ly9vbmVsYWtlLmJsb2IuZmFicmljLm1pY3Jvc29mdC5jb20vNTBmNWU2OTYtYzI0Ny00NzE4LTlhNjg
tYTkxMzIwNzM3Y2I0LzlmNWE3NmQ1LWI1OGMtNGEwOC04MjdhLWJlZjNiNThmYmFlYi9GaWxlcy9wcm9kdWN0X3Jldmlld3M
vZWxlY3Ryb25pY3MuanNvbg2",
  "product_category_name_english": "electronics",
  "review_count": 1168,
  "average_score": 3.6172945205479454,
  "min_score": 1,
  "max_score": 5,
  "reviews": [
    "The purchase was made easily. The delivery was made well before the given deadline. The pro
duct has already started being used and so far, no problems.",
    "I received only 1 Midea Split ESTILO controller. The remote control for Consul air conditio
ner was missing.",
    "I received only 1 Midea Split ESTILO controller. The remote control for Consul air conditio
ner was missing.",
    "Very good and fast when I saw it had already arrived",
    "Very good and fast when I saw it had already arrived",
    "Excellent service",
    "Great sale, everything perfect.",
    "I am still waiting for a response today about my products, they have already been paid for
and I have not received them yet, there were 3 identical ones and I need to speak with people, n
ot machines or recordings.",
    "I am still waiting for a response today about my products, they have already been paid for
and I have not received them yet, there were 3 identical ones and I need to speak with people, n
ot machines or recordings.",
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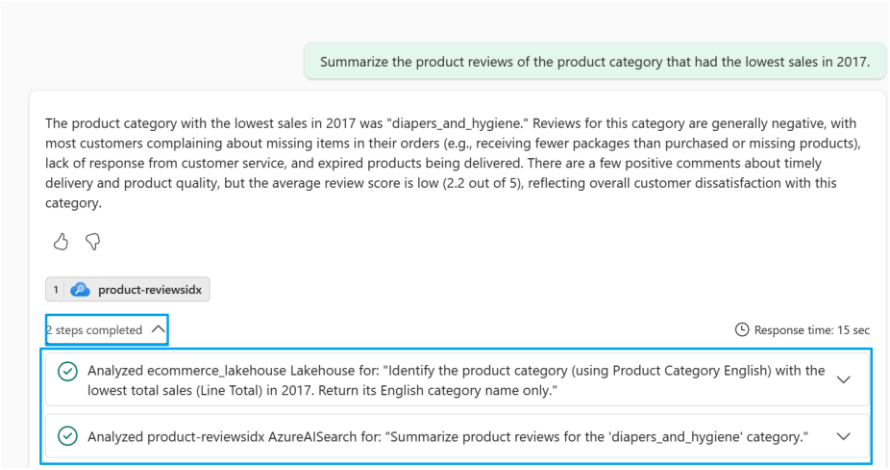
3. Now let's try a slightly more complex question that requires the Data Agent to reason across multiple data sources.

   Try asking "***Summarize the product reviews of the product category that had the lowest sales in 2017.***"



   From the response you can observe that the Data agent broke the question into two steps - first ecommerce lakehouse and then output to be queried against product-reviewsidx search index.

Congrats, you have officially finished the lab!